STM-GARSOS Processing: Detailed Methods

STM-GARSOS workflow

September 30, 2026

Reconstruction objective

- Dynamic image model: $X_t(x,y) = \sum_{\ell=1}^{L} \mathrm{ST}_{t,\ell}(x,y) C_\ell(x,y)$.
- Unknown: C_l(x,y), an [Nx x Ny x L] coefficient volume.
- Measured data: multicoil radial samples for each temporal frame.
- Goal: fit the measured radial data while restricting temporal variation to the STM subspace.

Entry point and inputs

- main_STM_GARSOS loads input k-space identifiers and scan metadata.
- A vendor-reconstructed reference volume defines target geometry.
- prepareGARSOSSTMData creates shared prepared data and calibration metadata.
- Historical defaults: 15 spokes/frame, 32 x 32 calibration square, L = 7.

Prepared trajectory and data

- sampledData: [nReadout x nPartitions x nSpokes, nCoils].
- kSpaceLocations: flattened [kx, ky, kz] rows.
- spokeIndex: one identifier per flattened sample.
- DCF: one radial density-compensation value per sample.
- Coordinates use normalized cycles/FOV-like units.

Trajectory construction

- $k_x = \rho\cos(\phi)$ and $k_y = \rho\sin(\phi)$.
- rho is the centered readout coordinate normalized by readout length.
- phi follows the GAR-SOS spoke-angle convention.
- kz is generated from the partition index and acquisition geometry.
- The arrays are flattened in readout/partition/spoke order.

Readout processing and whitening

- Readout delay correction is estimated from selected calibration lines.
- Readout Fourier processing is applied before the final flattening.
- Partition padding is performed during preparation.
- Noise covariance is estimated from noise acquisitions.
- The whitening transform is applied across receive channels.

Density compensation

- The existing DCF is based on radial shell-sector area.
- It is generated from the in-plane radius and readout increment.
- Frame-specific DCF values are selected with the frame samples.
- Calibration gridding applies DCF and additional reconstruction weights.
- The later STM radial operator uses its prepared radial samples and NUFFT convention.

Frame grouping

- Unique spoke identifiers define the acquisition order.
- nFrames = floor(nSpokes / nSpokesPerFrame).
- Each frame contains all readout and partition samples for its spokes.
- Incomplete trailing spokes are excluded.
- For the baseline, 15 spokes/frame produced approximately 133 frames.

Calibration reconstruction

- reconstructGARFrameForSTM selects frameData, frameTrajectory, and frameDCF.
- Coordinate transform and translation phase correction are applied.
- Gaussian/spatial weighting and DCF scaling are applied.
- GpuReconstructur20181008 grids every coil separately.
- The class applies inverse FFT, resize/deapodization, and coil-profile accumulation.

3-D grid and target geometry

- Target image geometry: imageSize = [224, 224, 96].
- Oversampled grid: gridSize = ceil(1.375 x imageSize) = [308, 308, 132].
- Samples are deposited with a 3-D Kaiser-Bessel kernel.
- ifftn is applied on the oversampled Cartesian grid.
- shiftAndResize maps the result to the centered 224 x 224 x 96 grid.

Calibration k-space extraction

- Select reconstructedVolume(:, :, zSlice).
- Apply fftshift(fft2(ifftshift(senseImage))).
- Extract the centered calibration square with localCentralRegion.
- Baseline calibrationSize = [32, 32].
- Single-slice kCal has dimensions [32 x 32 x nFrames].

STM computation

- `STM_computation` receives k_{Cal}, desired map dimensions, L, and options.
- The calibration data encode temporal correlations across frames.
- The algorithm estimates ST_maps [Nx x Ny x Nt x L].
- eigenValues provide a diagnostic of model rank/content.
- The baseline used L = 7 because more modes appeared informative than in the Cartesian demo.

Historical z-slice preparation

- prepareGARSOSSliceDataForMCNUFFT restores [readout, partition, spoke, coil].
- It applies the selected translation phase and coordinate transform.
- Historical baseline: centered direct IFFT along the 58 partition entries.
- The requested zIndex is selected from that 58-point result.
- This differs from the later 58-to-96 gridded z-only correction.

Radial STM data dimensions

- After frame grouping: radialData [nReadout x nSpokesPerFrame x nCoil x nFrame].
- kxy [nReadout x nSpokesPerFrame x nFrame].
- The in-plane radial trajectory is retained.
- The coil dimension is not combined before the STM solve.
- Trajectory/data correspondence is validated before operator construction.

Forward sensitivity-map model

- Forward coil model: $d_c = \mathcal{N}_{\mathrm{radial}}(S_c \odot X)$.
- S_c are forward receive sensitivities in the whitened channel basis.
- Adjoint coil combination is $\sum_c S_c^* d_c$ after NUFFT adjoints.
- This is distinct from calibration-only coil-combination profiles.
- Map and dynamic-data channel ordering must match.

Radial NUFFT operator

- buildGARSOSRadialEncodingOperator creates one NUFFT plan per frame.
- Forward: multiply by coil maps, then NUFFT each coil.
- Adjoint: NUFFT-adjoint each coil, then sum conjugated maps.
- Optional single precision converts trajectories, maps, data, and outputs.
- The operator exposes forward, adjoint, and vectorized forms.

STM operator composition

- buildGARSOSSTMOperators composes STM synthesis with radial encoding.
- $\mathcal{A}(C) = \mathcal{F}(\mathcal{S}(C))$.
- $\mathcal{A}^H(y) = \mathcal{S}^H(\mathcal{F}^H(y))$.
- The normal operator is applied without explicitly forming a matrix.
- The coefficient vector has Nx x Ny x L entries.

PCG solve

- Right-hand side: $b = \mathcal{A}^H y$.
- Unregularized system: $(\mathcal{A}^H \mathcal{A})C = b$.
- The baseline used tolerance 1e-5 and up to 50 iterations.
- Residual history, status flag, iteration count, and elapsed time are stored.
- Preconditioning and Toeplitz support were preliminary rather than baseline assumptions.

Optional Tikhonov solve

- Regularized system: $(\mathcal{A}^H\mathcal{A} + \lambda I)C = b$.
- The implementation can run a second PCG solve with lambda.
- The effective lambda depends on data, map, NUFFT, and DCF scaling.
- A numerical lambda should not be compared across differently scaled operators without calibration.

Synthesis and outputs

- $\widehat{X}_t = \sum_{\ell=1}^{L} \mathrm{ST}_{t,\ell} \widehat{C}_\ell$.
- sense_recon_stm has dimensions [Nx x Ny x nFrame].
- The result also contains coefficient images and PCG diagnostics.
- The initial reported output contained approximately 133 reconstructed frames.

Baseline versus later improvements

- Baseline: direct 58-point partition IFFT for STM slice data.
- Later work: forward sensitivity-map estimator and map-semantics audit.
- Later work: coil-preserving gridded z-only path for the 96-slice geometry.
- Later work: scaling, conditioning, coverage, regularization, and preconditioning analysis.
- These improvements should be presented separately from the historical baseline.

End-to-end summary

- Raw GAR-SOS data -> corrected, whitened radial samples.
- Trajectory + DCF + spoke IDs -> temporal frames.
- 3-D gridding -> calibration Cartesian k-space -> central square.
- STM calibration -> ST maps and eigenvalues.
- Coil-resolved radial data + sensitivity maps -> STM/radial operator.
- PCG -> spatial coefficients -> dynamic image series.